



JUNCTIONX KOREA 2026 · UPSTAGE STUDIO TRACK · TEAM 정선없음 (JEONGSYEON-EOPSEUM)

DocDo

Seniors photograph a letter and hear what it says. Their adult children see it checked against official records and approve. An agent completes the task. A person handles the authentication.

THE PROBLEM

For a low-literacy senior, mail is not a reading problem. It is a judgment problem.

What is this?

Bills, notices, collection letters and ads arrive in the same envelope. Reading the words does not tell you which one it is.

Is it real?

Impersonation letters keep the amount and due date intact and change only the phone number or the account.

What should I do?

"Just pay it" is the sentence that moves money. No one should have to decide that alone.

So the senior's screen only explains, reassures and warns. Every action moves to the child's screen.

Where aging arrived first, government still arrives on paper.

1 in 4

Gyeongbuk residents aged 65+: 26.1%, among the highest in Korea

49.2%

Uiseong County's 65+ share, the highest of all 226 districts

19

Gyeongbuk districts already super-aged, the most of any province

**16% →
30%**

Voice-phishing victims aged 60+, 2020 → H1 2025

The mailbox is the closest attack path

Local tax, water, health insurance, pension and welfare notices all come on paper. The regulator's newest top scam: fake "registered mail delivery" messages.

Family is far away

38.7% of households include an older adult. For a parent alone, a bill means photo → chat → a phone call → still no certainty.

Sources: Statistics Korea, 2025 Statistics on the Aged · Ministry of the Interior resident registration, Sept 2025 · Financial Supervisory Service

HOW IT WORKS

One photo. Read, verified, approved, executed.

Senior takes a photo No account. One link from the child, then a home-screen icon.	Upstage Studio Document Parse → Classify → Extract (two branches)	Verification layer Extracted facts checked against an official issuer registry	Two screens Senior: large text and voice. Child: verdict, key values, to-dos, evidence.	Solar agent Looks up the bill on the payment portal. Pays only if the amount matches.	Human turn Authentication is done by the child, by hand, on their phone.
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Never on the senior's screen

Account numbers, pay buttons, payment links, application buttons. The only tappable phone number is the issuer's official one from the registry.

When it is done

The senior's screen updates itself: "Handled · confirmation number". The loop closes.

Studio handles the document. Solar Pro 4 decides the next action.

Studio agent: Parse → Classify → Extract ×2

- Classify splits documents into **pay / apply / info / ad**. Only pay and apply reach Extract. Ads are intentionally unmapped.
- Calls `use include: [ "all" ]` to get per-field confidence and coordinates. A value marked `low` is never spoken or shown.
- If Classify's label is `low` but the score is ≥ 0.9 , extraction continues and the verification layer decides.

Solar Pro 4 browser agent

- Reads the page's accessibility tree (buttons, inputs, links with reference ids) and picks exactly one action: `click`, `type`, `press`, `scroll`, `wait_human`, `done`, `abort`.
- Not pixel computer-use. It never sees an image; it reads structure.
- Guardrails live in code: any password or certificate prompt hands over to a human before the model is called; pay buttons only when the document amount is visible; no navigation outside the allowed domain.

Platform findings we documented along the way: draft configs are invisible to the API · an unconfigured node blocks saving · `[ "last" ]` drops confidence · an unmapped type removes the Extract step from the response entirely.

The senior's screen never leads to money.

1

No path to a transaction: no accounts, no pay buttons, no payment links.

2

Uncertain values are not read. Extract low or missing means silence.

3

Facts, not instructions. Not "you should pay", but "the letter says 73,000 won".

4

Never tells anyone to throw a document away.

5

Contacts and links extracted from the letter are never actionable. Calls go only to registry numbers.

Wording rule

No verified mismatch is the strongest thing we say. Never "genuine", "normal" or "safe". The screen shows what was checked, how many, and what was not.

Agent rules

Never types passwords or certificates · aborts if the amount differs · stops after 3 identical screens or 25 steps · done only after the pay click.

We don't claim AI spots fakes. We compare against official records.

R1 Issuer match	Issuer name, phone, URL against the official registry. Department lines are accepted by area-code range.
R3 Confidence gate	Any low or missing key field: numbers are not read aloud.
R4 Personal mobile	A public agency bill with a 010 contact number is a hard warning.
R5 Classification	If the document type is uncertain, a person decides.
R6 Account holder	The virtual-account holder must equal the issuer name exactly.

Controlled experiment: two fields changed		
Field	Control	Tampered
Contact phone	1577-1000	1588-9876
Website	nhis.or.kr	nhis-pay.co.kr
Amount, due, issuer	Identical. Extract read both correctly.	
Verdict	clear 2/2	mismatch 0/2
The verdict was decided by the registry check, not by the model.		

Numbers are reported as observed ranges.

12 / 12

Team-made fixtures, expected verdict

12 · 11 / 12

Photo-degradation simulation, mid · hard. The hard miss went low and was handed to a person.

296

Regression tests, including 19 attack and missing-input cases and bypasses reproduced in review

4.1–26.2 sec

Processing time, observed range, n = 12. Not a P95.

Three printed letters today, real photos

Busan Waterworks no verified mismatch ·

"Seoul Waterworks" (fake) suspected impersonation ·

Busan District Court needs review

Registry

Pilot coverage of 8 issuers (NHIS, MOHW, NPS, KEPCO, Pohang City, Busan District Court, Seoul Arisu, and a demo entry for Busan Waterworks). Private companies are not checked.

When input degrades, it does not fail quietly. It hands over.

The child's turn: a live screen they can tap on their phone.

1 · Approve

Child's card: [Approve payment · 12,500 won]

2 · Look up

Chromium worker on Railway.
Solar reads the structure and acts.

3 · Amount guard

Portal amount must equal the document amount.

4 · Human turn

Secure keypad, tapped by the child on the live frame.

5 · Done

Receipt → senior's screen says "Handled".

Why hand over

Authentication, passwords and consent are legal acts. The agent stops the moment it sees them and resumes only after the child presses [Continue]. Remote inputs pass through a queue, are never stored, and are cleared when the run ends.

Demo portal

The real Korean giro site requires a financial certificate even for guests, so lookups sit behind authentication. We run the same flow on a demo portal (always labelled "demo · no real payment") and show the real site up to its authentication wall. Connecting a real issuer means swapping the ledger for an API, not rewriting the agent.

Only one thing changes: who sits in the child's seat.

1 · Care workers' case management

A care worker photographs the mail on a visit; today's child screen becomes their case screen. Who got which bill, due when, handled or not, on record.

3 · A Gyeongbuk issuer registry

Register official numbers and accounts for local tax and water bills across 23 cities and counties, and every public notice in Gyeongbuk becomes checkable. Data entry, not model work. Flagged patterns go back to local government and police.

2 · Township offices and senior centers

Seniors without a phone use the same flow on a counter or senior-center tablet. The senior side has no account, so any device works.

4 · Public payment APIs

When Wetax or Giro open an API, we swap the ledger. The agent stays the same. A human still authenticates.

Helps the family when there is one. Lets the public sector take that seat when there isn't.

What these results do not say

Not detected

- An impersonation letter that copies official contacts verbatim. Every value it carries is real, so a registry check passes it.
- Sub-office names (e.g. "NPS Pohang Branch") cannot be told apart from fakes by structure alone. Both go to the child for review. We accept false positives to reduce misses.
- We do not look up the real holder of a bank account. No authority. The screen says "not checked".

Next

- Web push for the child: "a letter arrived", "your turn".
- Registry growth from open government data, including local area-code ranges.
- Real issuer APIs behind the same agent.
- Show the child the original region of the photo using Extract coordinates, instead of reading the value.



DOCDO · "DOC DO": THE DOCUMENT ACTS

Seniors listen. Children decide. DocDo does the rest.

Demo docdo.vercel.app · Source github.com/needsbuilder/docdo (fully open, licenses in README)

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